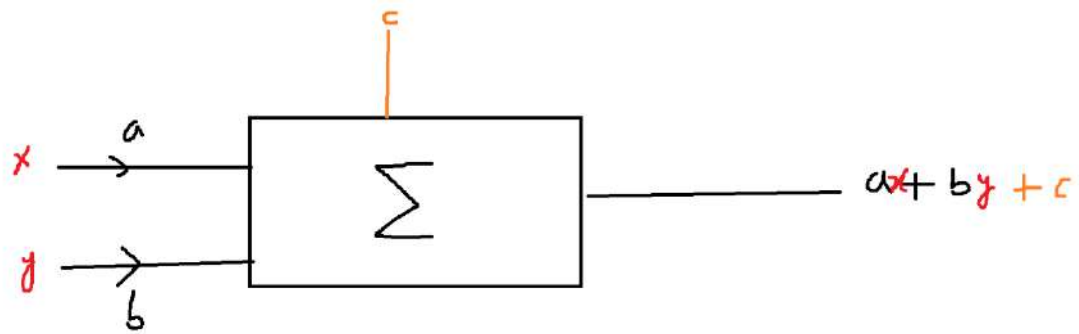




Logistic regression

$$p(y) = \frac{1}{1 + e^{-(b_0 + b_1 x_1 + b_2 x_2)}}$$

Step – 1: X_1 and X_2 are passing

Step – 2: X_1 is multiply with b_1 , X_2 multiple with b_2

Step – 3: b_0 is adding

*by this a equation: $y = b_0 + b_1 * x_1 + b_2 * x_2$*

Step – 4: the y is passing through Activation function or sigmoid functon

$$\sigma(y)$$

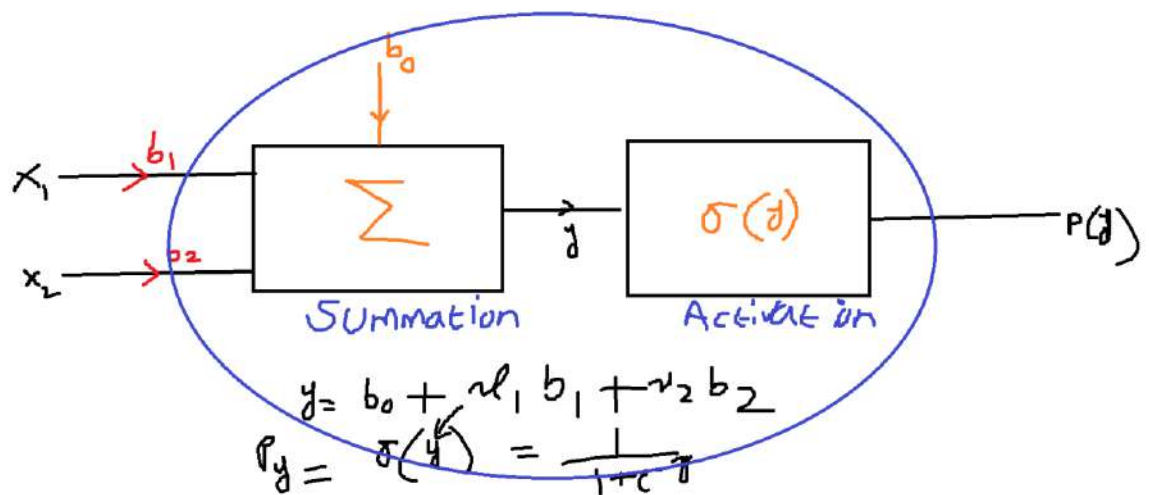
Step – 5 = $p(y)$

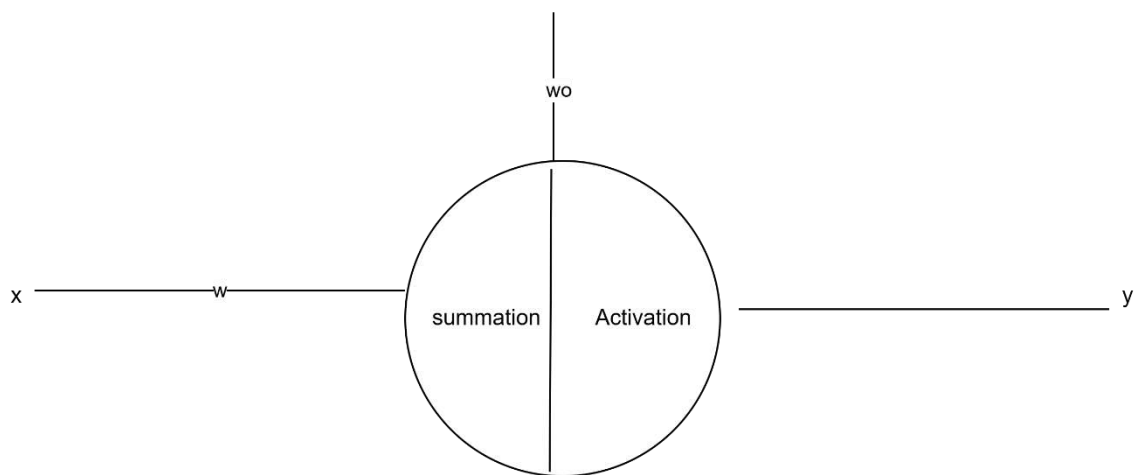
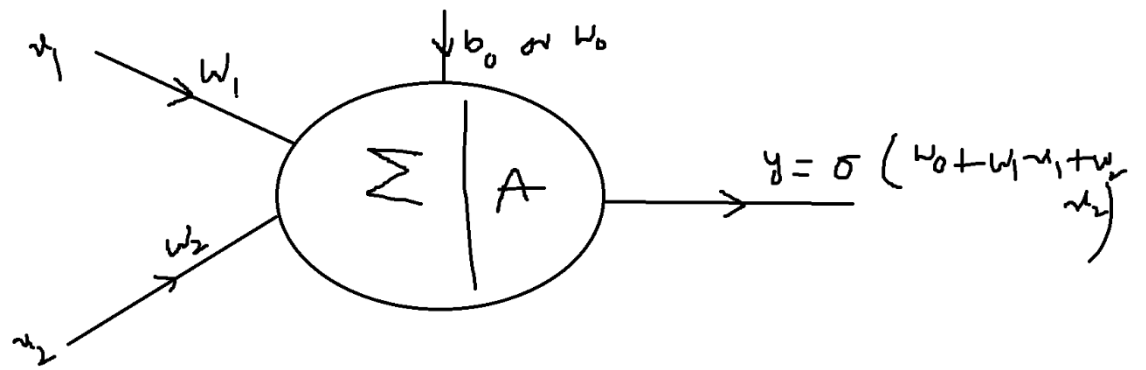
Nuerorn:

Nuerorn:

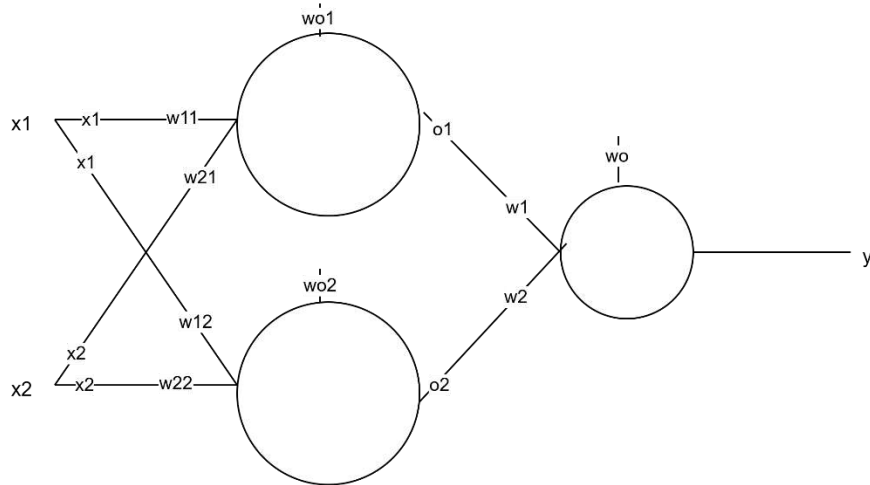
Deep learning consit of Nuerons
each neuron represnts with a round circle
In each neuron there are two operations

- summation
- Activation





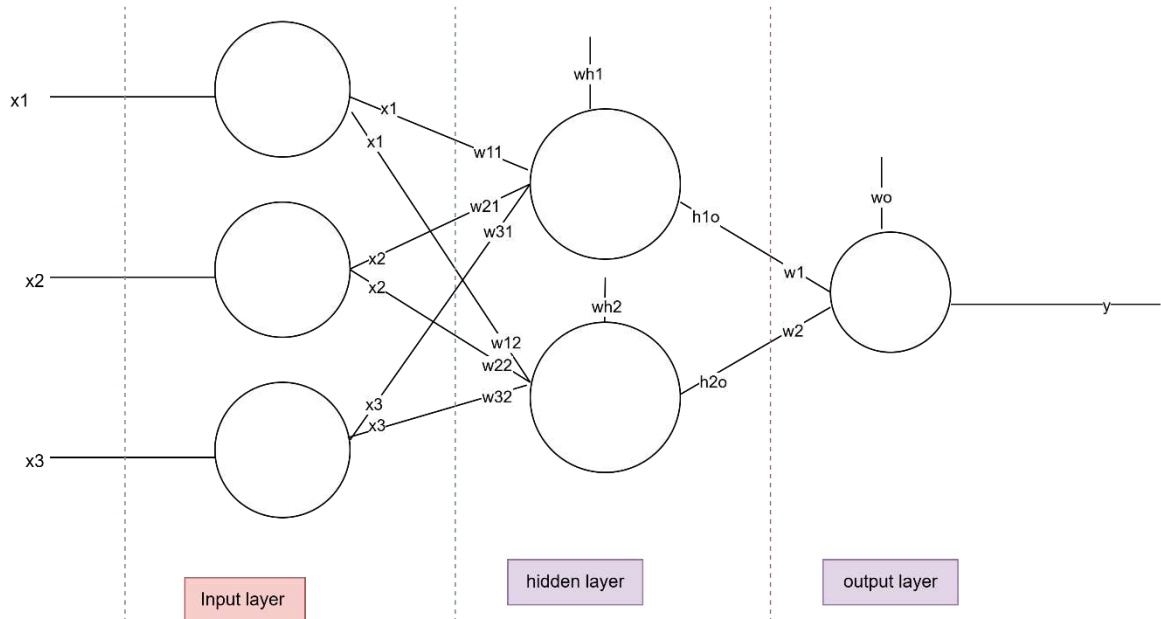
$$Y = \text{Activation}(w_0 + wx)$$



$$y = \text{Activation}(wo + w_1 * o_1 + w_2 * o_2)$$

$$o_1 = \text{Activation}(wo_1 + x_1 * w_{11} + x_2 * w_{21})$$

$$o_2 = \text{Activation}(wo_2 + x_1 * w_{12} + x_2 * w_{22})$$



$$y = \text{activation}(w_o + w_1 * h_{1o} + w_2 * h_{2o})$$

$$h_{1o} = \text{activation}(w_{h1} + w_{11} * x_1 + w_{21} * x_2 + w_{31} * x_3)$$

$$h_{2o} = \text{activation}(w_{h2} + w_{12} * x_1 + w_{22} * x_2 + w_{32} * x_3)$$

Three types of layers available

- 1) Input layer*
- 2) Hidden layer*
- 3) Output layer*

Input layer

- *Input layer is used to pass the inputs*
- *Input layer consist neurons*
- *The number of neurons equal to number of inputs*
- *For example in wine qulaity datasets the number inputs = number of columns*
- *there are 11 input columns are there so we have 11 neurons*
- *For example an image size has $28 * 28 = 784$ pixels we have*
- *so that we required 784 neurons*
- *Input neurons used only for passing the inputs*
- *While passing it will not multiply with any weights*
- *So in the input neurons no summation and no activation*

Hidden layers

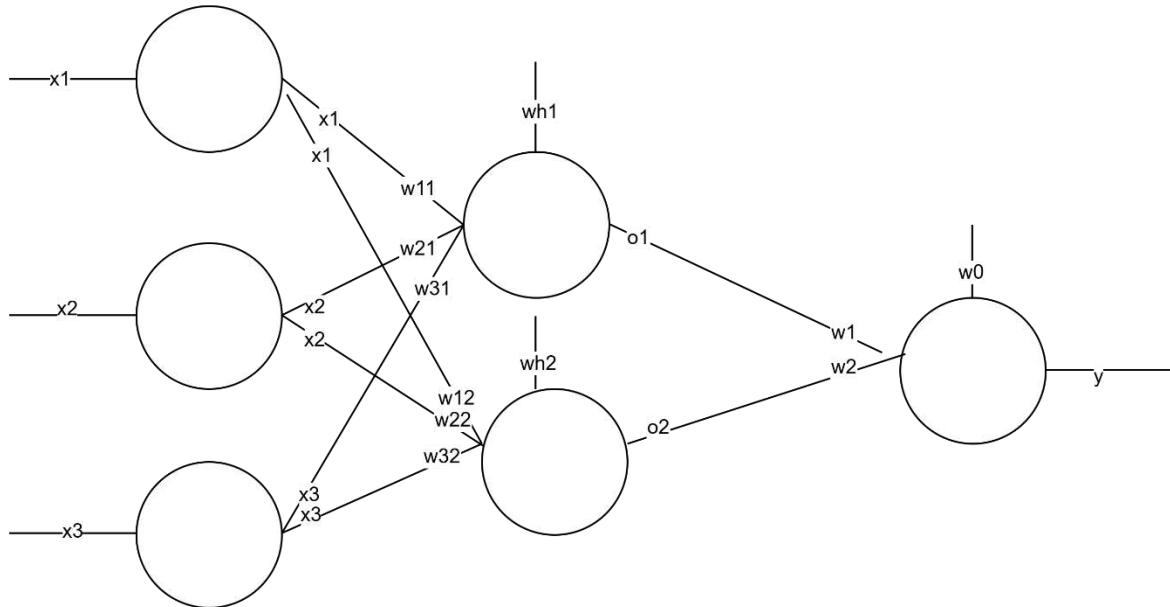
- *Hidden layers are between input and output layer*
- *It is called hidden because it does not takes the inputs directly*
- *The input value for hidden layer is: input data multiply with weight*
- *hidden layers are used to identify the patterns*

Output layer

- *Output layer is used to provide outputs*
- *Bi classification : One neuron*
- *Multi classification: Multiple neurons*
- *10 classes : 10 neurons*
- *Each neuron has bias and weights*

Case – 1 We have seen the Network

Input layer(3) – Hidden layer(2) – output layer(1)



$$y = \text{activation}(w_0 + o_1 * w_1 + o_2 * w_2) = 3$$

$$o_1 = \text{activation}(w_{h1} + x_1 * w_{11} + x_2 * w_{21} + x_3 * w_{31}) = 4$$

$$o_2 = \text{activation}(w_{h2} + x_1 * w_{12} + x_2 * w_{22} + x_3 * w_{32}) = 4$$

$$\text{Total parameters} = 3 + 4 + 4 = 11$$

layer(nuerons)	Parameter(weights + bias)
Input layer (3)	0
Hidden layer(2)	$3 * 2 + 2 = 8$
output layer(1)	$2 * 1 + 1 = 3$
	11

Case – 2 We have seen the Network

Input layer(4) – $H_1(3)$ – $H_2(2)$ – output layer(1)

Input layer = 4 neurons

hidden layer1 = 3 neurons

hidden layer2 = 2 neurons

output layer1 = 1 neurons

we can have many hidden layers between input and output

The more we use hidden layers that is called as Deep Neural Network (DNN)

Two qns arise

- *We know input layer always one layers but many neurons*
 - *The number of neurons in input layer = Number of inputs*
- *We know only one output layer available:*
 - *The number of neurons in output layer*
 - *Bi classification: 1 neuron*
 - *Multi classification = 10 classes = 10 neurons*

How may Hidden layers we need to choose?

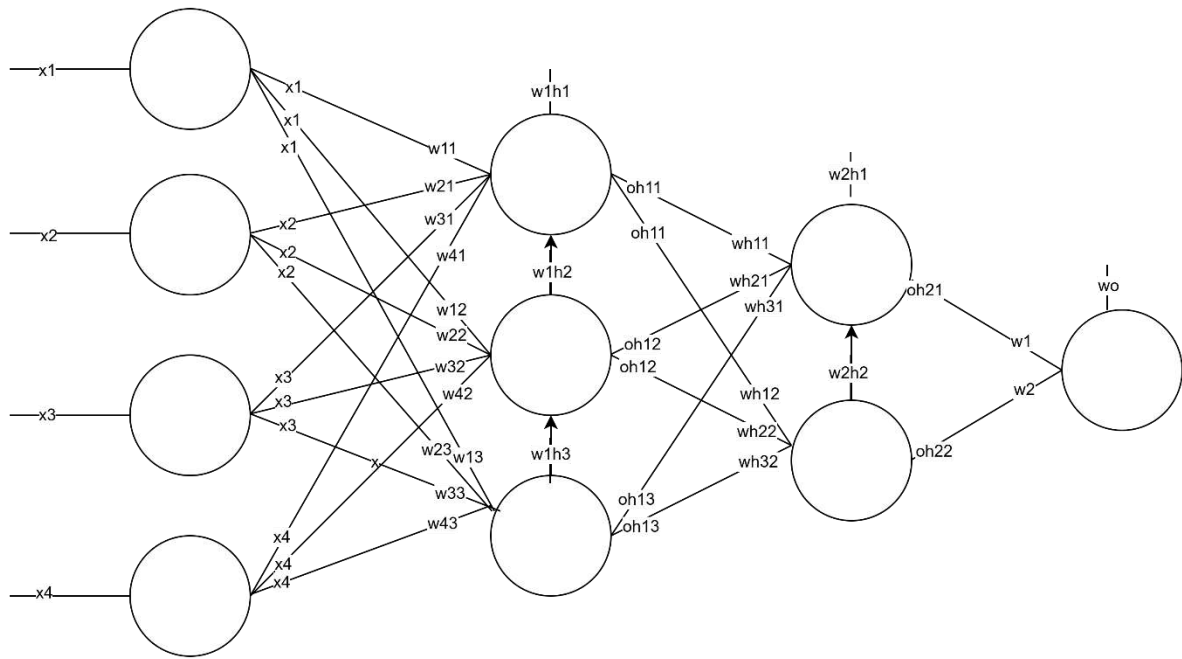
How many neurons in each hidden layer?

Are you able to draw NN

Are you able to write maths NN

Are you able to calculate parameters

Input layer(4) – $H_1(3)$ – $H_2(2)$ – output layer(1)



oh_{11} = output of first hidden layer – first neuron

oh_{12} = output of first hidden layer – second neuron

oh_{13} = output of first hidden layer – Third neuron

wh_{11} = weight between hidden layer first neuron to first neuron

wh_{12} = weight between hidden layer first neuron to second neuron

$$y = \text{activation}(w_o + oh_{21} * w_1 + oh_{22} * w_2) = 3$$

$$oh_{21} = \text{activation}(w_{2h1} + oh_{11} * wh_{11} + oh_{12} * wh_{21} + oh_{13} * wh_{31}) = 4$$

$$oh_{22} = \text{activation}(w_{2h2} + oh_{11} * wh_{12} + oh_{12} * wh_{22} + oh_{13} * wh_{32}) = 4$$

$$oh_{11} = \text{activation}(w_{1h1} + x_1 * w_{11} + x_2 * w_{21} + x_3 * w_{31} + x_4 * w_{41}) = 5$$

$$oh_{12} = \text{activation}(w_{1h2} + x_1 * w_{12} + x_2 * w_{22} + x_3 * w_{32} + x_4 * w_{42}) = 5$$

$$oh_{13} = \text{activation}(w_{1h3} + x_1 * w_{13} + x_2 * w_{23} + x_3 * w_{33} + x_4 * w_{43}) = 5$$

$$\text{Total parameters} = 3 + 4 + 4 + 5 + 5 + 5 = 26$$

layer(nuerons)	Parameter(weights + bias)
----------------	---------------------------

<i>Input layer (4)</i>	0
<i>Hidden layer(3)</i>	$4 * 3 + 3 = 15$
<i>Hidden layer(2)</i>	$3 * 2 + 2 = 8$
<i>output layer(1)</i>	$2 * 1 + 1 = 3$
	26

Case – 3

Input layer(5) – $H_1(4)$ – $H_2(3)$ – $H_3(2)$ – output layer(1)

Input layer = 5 neurons

Hidden layer₁ = 4 neurons

Hidden layer₂ = 4 neurons

Hidden layer₃ = 2 neurons

Output layer = 1 neuron

<i>layer(nuerons)</i>	<i>Parameter(weights + bias)</i>
<i>Input layer (5)</i>	0
<i>Hidden layer₁(4)</i>	$5 * 4 + 4 = 24$
<i>Hidden layer₂(4)</i>	$4 * 4 + 4 = 20$
<i>Hidden layer₃(2)</i>	$4 * 2 + 2 = 10$
<i>output layer(1)</i>	$2 * 1 + 1 = 3$
	57

Case – 5

Input layer(10) – $H_1(8)$ – $H_2(6)$ – $H_3(4)$ – $H_4(2)$ – output layer(1)

Input layer = 10 neurons

Hidden layer₁ = 8 neurons

Hidden layer₂ = 6 neurons

Hidden layer₃ = 4 neurons

Hidden layer₄ = 2 neurons

Output layer = 1 neuron

$$10 * 8 + 8 * 6 + 6 * 4 + 4 * 2 + 2 * 1 = 80 + 48 + 24 + 8 + 2 = 162 \text{ weights}$$

$$8 + 6 + 4 + 2 + 1 = 21 \text{ Bias}$$

$$162 + 21 = 183$$

Case – 6

Input layer(108) – H₁(28) – H₂(21) – H₃(10) – H₄(7) – output layer(10)

4038

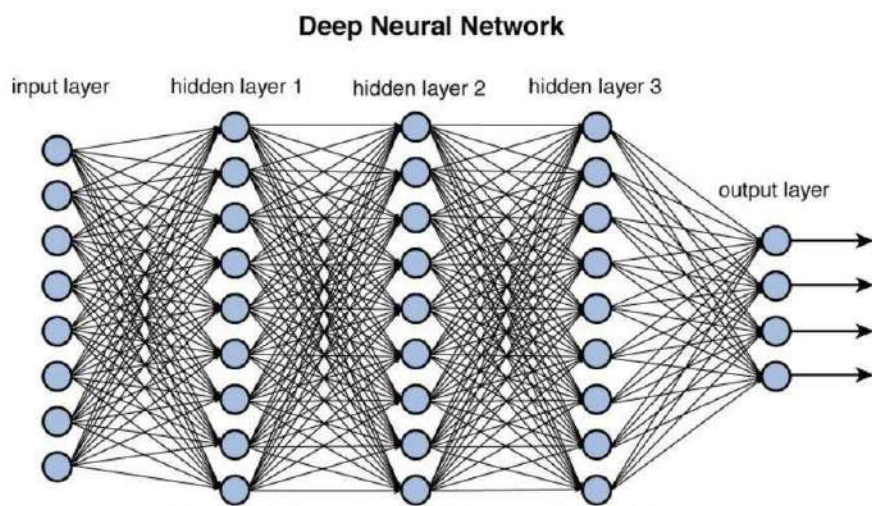


Figure 12.2 Deep network architecture with multiple layers.

*Input image is 28 * 28 shape ===== Input neuron = 784*

hidden layer = 128

output layer = 10

$$784 * 128 + 128 = 100480$$

$$128 * 10 + 10 = 1290$$

101770